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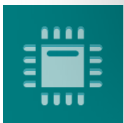
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Autonomous driving systems face critical perception failures in dense fog, where conventional RGB cameras suffer from severe degradation due to atmospheric scattering and reduced visibility. This paper presents an adaptive multi-modal fusion framework that synergistically integrates gated imaging with 3D LiDAR point clouds to achieve robust obstacle detection under visibility conditions as low as 50 m. Unlike standard cameras that passively capture scattered ambient light, gated cameras employ time-synchronized active illumination to physically filter backscattered photons, preserving structural features even in low-visibility scenarios. We propose a novel Adaptive Feature-Weighting Network (AFW-Net) that dynamically adjusts sensor modality contributions based on real-time environmental degradation assessment. The framework incorporates three key innovations: (1) a cross-modal feature extraction module that exploits the complementary physical properties of gated imaging and LiDAR, (2) an attention-based adaptive fusion mechanism that quantifies per-modality reliability through uncertainty estimation, and (3) a dynamic sensor scheduling module that optimizes resource allocation. Extensive experiments on the FogBank dataset demonstrate that our approach achieves a 23% improvement in detection accuracy (50 m visibility) compared to baseline methods, validating the necessity of the proposed AFW-Net for robust perception in adverse conditions. This work provides a new paradigm for safety-critical autonomous systems operating in low-visibility environmental conditions.

Keywords: gated imaging (/search?q=gated+imaging); LiDAR (/search?q=LiDAR); adaptive sensor fusion (/search?q=adaptive+sensor+fusion); uncertainty estimation (/search?q=uncertainty+estimation); 3D object detection (/search?q=3D+object+detection); adverse weather (/search?q=adverse+weather); dense fog (/search?q=dense+fog); autonomous driving (/search?q=autonomous+driving)

1. Introduction

The rapid advancement of autonomous driving systems has positioned perception as the cornerstone of safe and reliable vehicle operation. Modern autonomous vehicles rely heavily on multi-modal sensor suites, typically comprising RGB cameras, LiDAR, and radar, to construct

comprehensive environmental representations [1]. However, the performance of these perception systems degrades significantly under adverse weather conditions such as dense fog, heavy rain, and snow, where atmospheric scattering and particle occlusion severely compromise sensor reliability [2,3]. Studies have shown that conventional RGB cameras suffer from exponential visibility degradation in fog with meteorological optical ranges below 100 m [3], while LiDAR sensors experience substantial point cloud sparsification and noise amplification due to raindrop reflections and snowflake interference [4,5].

The safety-critical nature of autonomous driving mandates robust perception capabilities across all operational conditions. According to transportation safety statistics, adverse weather conditions contribute to approximately 21% of vehicle crashes and 16% of crash fatalities annually in the United States alone [6]. Current state-of-the-art perception systems, predominantly based on RGB-LiDAR fusion architectures [7,8], exhibit pronounced performance degradation when visibility drops below 50 m—a common occurrence in dense fog or heavy precipitation scenarios. This phenomenon, often characterized by a cliff-edge performance drop rather than a gradual decline, poses a significant challenge for autonomous driving systems.

Recall that in adverse weather conditions, the clear and accurate perception of the environment is crucial for autonomous driving. [9]. State-of-the-art LiDAR perception systems struggle to recover performance in these scenarios. When perception is compromised, the system's ability to process and interpret the environment is significantly reduced.

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business in between training sets from attempt to sorption. al post-

Gated imaging technology addresses these physical limitations of passive sensing from a different angle. Unlike conventional cameras that continuously integrate all incoming photons, gated cameras employ time-synchronized active illumination paired with nanosecond-precision electronic shuttering [2,11]. This temporal gating mechanism enables selective capture of photons that have traveled the direct path from illumination source to target and back, while rejecting the majority of scattered photons from atmospheric particles. The physical principle underlying this capability is as follows: direct-path photons arrive earlier than multiply scattered photons due to their shorter optical path length. By opening the camera shutter only during a narrow temporal window corresponding to a specific distance range, gated imaging suppresses backscatter at the hardware level, preserving scene structure even in dense fog with visibility as low as 30–50 m.

Despite the promising physical properties of gated imaging, several challenges impede its direct adoption in autonomous driving systems. First, gated cameras require active illumination, which introduces range limitations and potential interference in multi-vehicle scenarios. Second,

the temporal gating process captures only a depth slice of the scene per frame, necessitating multiple acquisitions or sophisticated reconstruction algorithms for complete scene coverage [11]. Third, and most critically, no single sensor modality provides optimal performance across all weather conditions—gated imaging excels in fog but offers no inherent advantage in clear weather, while LiDAR maintains geometric accuracy in many conditions but struggles with precipitation [12,13].

The key premise of this work is that robust all-weather perception requires adaptive fusion that dynamically adjusts sensor contributions based on real-time environmental conditions. Rather than treating different modalities as equally reliable throughout all scenarios, a fusion model should allocate greater weight to the sensors whose features are most reliable given the current degradation state. This principle motivates our proposed Adaptive Feature-Weighting Network (AFW-Net), which learns to assess per-modality reliability through uncertainty estimation and adjusts fusion weights accordingly.

We position our contribution within the body of confidence-aware and uncertainty-aware sensor-fusion research, which has been largely limited to static fusion weights or condition-dependent fusion weights. Our work introduces a learned, uncertainty-driven adaptive weighting mechanism that dynamically rebalances sensor contributions at the feature level; to our knowledge, no prior gated-LiDAR fusion work provides this capability. Second, we design a degradation-aware training strategy that implicitly teaches the network to recognize weather-induced feature corruption through synthetic augmentation, eliminating the need for explicit weather condition labels at inference time. Third, we provide a systematic empirical study indicating that adaptive gated-LiDAR fusion maintains robust performance under dense fog

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conditions where RGB-LiDAR methods suffer severe failure, achieving over 23% improvement in detection precision on the evaluated dataset.

Our approach addresses three research questions: (1) How can we effectively extract and fuse complementary features from gated images and LiDAR point clouds given their disparate data structures and physical sensing principles? (2) How can the system autonomously assess the reliability of each sensor modality in real-time without explicit weather condition labels? (3) How can we ensure that the fusion mechanism gracefully transitions between modality-dominant regimes rather than exhibiting discontinuous behavior?

To answer these questions, we make the following contributions:

- We propose a cross-modal feature extraction architecture that leverages specialized encoders for gated imaging and LiDAR while establishing geometric correspondence through camera-LiDAR projection. The gated image encoder employs a residual architecture with channel attention to capture both intensity and range information encoded in gated acquisitions, while the LiDAR encoder utilizes PointNet++ [16] with semantic abstraction to extract hierarchical geom
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- degradation patterns and environmental conditions without requiring explicit weather annotations during inference.
 - We conduct experiments on the Princeton Automated Driving Dataset, which contains synchronized gated imaging, LiDAR, and GPS data captured under diverse weather conditions. On this dataset, AFW-Net maintains average precision above 82% under dense fog conditions (50 m visibility), outperforming RGB-LiDAR fusion baselines that degrade to 58.4% AP under the same conditions.

The remainder of this paper is organized as follows. **Section 2** reviews related work in multi-modal fusion for autonomous driving, adverse weather perception, and gated imaging technologies. **Section 3** provides necessary background on gated imaging principles and problem formulation. **Section 4** details the proposed methodology, including network architecture, adaptive fusion mechanism, and training strategy. **Section 5** presents the

experimental results, ablation studies, and comparative analysis. Finally, **Section 6** concludes the paper with discussion of limitations and future research directions.

2. Related Works

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This section reviews prior research across three interconnected domains: multi-modal sensor fusion for 3D object detection, perception systems under adverse weather conditions, and gated imaging technologies for autonomous driving applications.

2.1. Multi-Modal Fusion for 3D Object Detection

Multi-modal fusion has emerged as a dominant paradigm in autonomous driving perception, leveraging complementary strengths of different sensor modalities to achieve robust 3D object detection. Early fusion approaches directly concatenate features from different modalities at the input level [17], while late fusion methods combine detection results from independent single-modality networks [18]. However, these naive fusion strategies fail to capture cross-modal correlations.

Recent level into point clouds a region LiDAR point detector voxel-based convolutional

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More sophisticated fusion architectures exploit attention mechanisms to model cross-modal dependencies. SECOND [21] introduced sparse convolutional networks for efficient 3D detection from point clouds, while PointPillars [22] simplified the representation by encoding point clouds into vertical pillars. Building upon these foundations, CLOCs [23] and 3D-CVF [24] developed cross-modal alignment modules that establish dense correspondences between image pixels and 3D points. PointAugmenting [25] proposed a decorrelated feature augmentation strategy to enhance cross-modal feature learning.

Transformer-based architectures have recently gained traction for multi-modal fusion. TransFusion [14] employs cross-attention mechanisms to fuse image and LiDAR features in a unified framework. BEVFormer [15] constructs bird's-eye view representations from multi-camera images using spatial and temporal transformers, representing a modern trend toward unified BEV-based perception architectures rather than traditional multi-modal fusion. However, these methods predominantly assume consistent sensor reliability across operating conditions,

making them vulnerable to performance degradation when individual modalities fail under adverse weather.

A related line of research explicitly models predictive uncertainty to improve fusion robustness. Confidence-aware and probabilistic fusion methods estimate per-detection or per-region uncertainty and use it to weight or gate sensor contributions, drawing on aleatoric and epistemic uncertainty formulations from Bayesian deep learning. Most of these approaches, however, estimate uncertainty at the detection-output level and target average-condition calibration. In contrast, our method estimates uncertainty at the feature level for each modality and uses it directly to drive the fusion weights, with the estimator supervised against clean-condition feature targets so that the uncertainty has a concrete operational meaning under degradation (**Section 4.3** and **Section 4.4**). This distinction is what allows the fusion weights to shift systematically as a modality degrades, rather than only recalibrating the final confidence scores.

2.2. Perception Under Adverse Weather Conditions

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and ranging error. Kutila et al. [4] performed controlled experiments demonstrating that heavy rain can reduce LiDAR effective range by up to 50%, while Filgueira et al. [12] analyzed point cloud density degradation as a function of weather severity. These observations motivated the development of snow and rain removal algorithms. Kurup and Bos [10] proposed DSOR, a statistical filter that identifies and removes snowflake-induced noise points based on spatial-temporal consistency. Hahner et al. [5] introduced LiDAR snowfall simulation techniques to augment training data, demonstrating improved detection robustness through synthetic weather augmentation.

Domain adaptation techniques have been explored to bridge the distribution gap between clear and adverse weather conditions. WEDGE [27] constructed a multi-weather autonomous driving dataset using generative vision-language models, facilitating training under diverse weather conditions including fog, rain, and snow. Rothmeier and Huber [28] developed standardized test methodologies for evaluating vision-based detection algorithms under adverse

MDPI (0) conditions, while Hasirlioglu et al. [29] proposed sensor performance evaluation frameworks specific to rain scenarios.

Despite these advances, existing approaches face fundamental limitations when dealing with extreme degradation. Statistical filtering methods can remove weather-induced noise but cannot recover lost geometric information when point cloud density falls below critical thresholds. Domain adaptation relies on sufficient feature preservation in degraded images, which breaks down when visibility approaches sensor physical limits. These limitations motivate the exploration of alternative sensing modalities that maintain information content under severe weather conditions.

From a broader machine learning perspective, perception under extreme weather is also closely related to rare-event prediction in imbalanced, high-risk environments. Abdulrazaq [30] highlights that, in such scenarios, model development should be accompanied by evaluation protocols that emphasize minority and failure-critical cases, including precision–recall-oriented analysis and robustness-focused assessment rather than overall accuracy. This viewpoint aligns

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scattering phenomena such as fog and rain.

Bijelic et al. [2] demonstrated the first deep learning framework for gated imaging in adverse weather, showing that gated images preserve structural features even in dense fog with visibility below 50 m. Their multi-modal fusion network combined gated, RGB, and LiDAR inputs, achieving significant improvements over conventional sensors in synthetic and real foggy conditions. Gruber et al. [31] introduced Gated2Depth, a method to reconstruct dense depth maps from gated image sequences by leveraging the implicit range information encoded in gated acquisitions. Their approach employed adversarial training to handle the domain shift between gated and conventional imaging.

Subsequent work by Walia et al. [11] proposed self-supervised depth estimation from gated images without requiring LiDAR ground truth, using temporal consistency across multiple gated slices as the supervisory signal. This advancement reduced the data annotation burden for training gated image processing networks. Laurenzis et al. [32] developed physics-based

models for gated imaging in scattering media, providing theoretical foundations for understanding backscatter suppression as a function of gate timing parameters.

Despite these advances, several challenges hinder the widespread adoption of gated imaging in autonomous vehicles. First, active illumination requirements limit effective operating range and introduce potential interference in multi-vehicle scenarios. Second, gated cameras typically acquire depth slices sequentially, requiring either multiple gates per frame or sophisticated depth reconstruction algorithms. Third, gated imaging provides minimal advantage over conventional cameras in clear weather, motivating the need for adaptive fusion strategies that dynamically leverage gated images only when environmental conditions warrant their use.

Our work builds upon these foundations by introducing an adaptive fusion framework that treats gated imaging and LiDAR as complementary modalities with weather-dependent reliability. Unlike prior approaches that use fixed fusion weights [2] or modality-specific networks without cross-modal adaptation, we propose a unified architecture that learns to assess per-modality confidence and dynamically adjusts fusion weights based on real-time feature quality assessment.

3. Preliminaries

This section introduces the physical principle of gated imaging.

3.1. Gated Imaging

Gated imaging is a technique where the camera shutter is controlled to capture light only during a specific time interval (the gate).

The system uses a camera equipped with a fast electronic shutter. The operational sequence proceeds as follows:

1. A laser pulse of duration $\tau_p \approx 10$ ns illuminates the scene at time t_0 .
2. Photons propagate to objects at distance d , requiring time $t_{direct} = 2d/c$ for the round trip, where c is the speed of light.
3. Scattered photons from atmospheric particles travel longer optical paths, arriving with delays $t_{scatter} > t_{direct}$.
4. The camera shutter opens at time $t_{gate} = t_0 + t_{delay}$ for duration $\tau_g \approx 10$ ns, capturing only photons from the range slice $[d_{min}, d_{max}]$.

The range selectivity is determined by

$$d_{min} = \frac{c \cdot t_{delay}}{2}, \quad d_{max} = \frac{c \cdot (t_{delay} + \tau_g)}{2}. \quad (1)$$

This temporal gating effectively filters backscatter because multiply scattered photons from nearby particles accumulate additional optical path length, causing them to arrive outside the narrow temporal window. The benefit of gating can be expressed in terms of the signal-to-backscatter ratio (SBR). For a continuous-wave camera, backscatter is integrated over the full exposure time τ_{exposure} (on the order of milliseconds), whereas a gated camera integrates backscatter only over the short gate window τ_g (on the order of nanoseconds) that brackets the target range. Because the in-gate signal energy from the target is approximately preserved while the integrated backscatter scales with the integration window, the relative backscatter contribution is reduced by a factor of approximately $\tau_g / \tau_{\text{exposure}}$, so that

$$\text{SBR}_{\text{gated}} \approx \frac{\tau_{\text{exposure}}}{\tau_g} \cdot \text{SBR}_{\text{conventional}}, \quad (2)$$

i.e., the gated SBR improves over the conventional SBR by the ratio $\tau_{\text{exposure}} / \tau_g \gg 1$. With τ_{exposure} in the millisecond range and $\tau_g \approx 10$ ns, this improvement factor can exceed two orders of magnitude [2]. We note that this improvement is only realized if the backscatter is correctly modeled.

3.2. Problem Formulation

We consider a scene $\mathcal{S}$ represented by a set of 3D boxes $\mathcal{B}$ in a 3D coordinate system $\mathcal{C}$.

$\mathcal{B} = \{b_j\}_{j=1}^{N_l}$, where each box $b_j = (x, y, z, l, w, h, \theta, c)$ specifies the object's center position, dimensions, orientation, and class.

The key challenge is that sensor reliability varies with environmental conditions. Let $\eta_g(t)$ and $\eta_l(t)$ denote the time-varying reliability (information content) of gated imaging and LiDAR modalities at time t . Under clear conditions, $\eta_g \approx \eta_l$, but in dense fog, the reliability of conventional RGB imaging collapses ($\eta_{\text{rgb}} \ll \eta_l$) while gated imaging maintains high η_g . An optimal fusion strategy should satisfy

$$\mathcal{F}^*(\mathcal{G}, \mathcal{D}) = \arg \max_{\mathcal{F}} \mathbb{E} [\text{AP}(\mathcal{F}(\mathcal{G}, \mathcal{D}), \mathcal{B}_{\text{gt}}) \mid \eta_g(t), \eta_l(t)], \quad (3)$$

where AP denotes average precision and $\mathcal{B}_{\text{gt}}$ are ground-truth annotations.

Our approach models this through learned adaptive weights $w_g, w_l \in [0, 1]$ such that $w_g + w_l = 1$, where the weights automatically adjust based on implicit estimation of η_g and η_l .

from feature-level uncertainty. This formulation avoids requiring explicit weather condition labels during inference while enabling smooth transitions between modality-dominant regimes.

3.3. Feature Space Alignment

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A critical challenge in fusing gated images and LiDAR point clouds lies in their disparate data structures: images are dense 2D grids while point clouds are sparse 3D irregular sets. We establish correspondence through perspective projection. Let a 3D point be expressed in homogeneous LiDAR coordinates as $\mathbf{X} = [x, y, z, 1]^T$ and its image projection in homogeneous pixel coordinates as $\tilde{\mathbf{x}} = [u, v, 1]^T$. The projection is

$$s[u, v, 1]^T = K [R | t] [x, y, z, 1]^T, \quad (4)$$

where $K \in \mathbb{R}^{3 \times 3}$ is the camera intrinsic matrix, $[R | t] \in \mathbb{R}^{3 \times 4}$ is the extrinsic transformation (rotation $R \in SO(3)$ and translation $t \in \mathbb{R}^3$) from LiDAR to camera coordinates, and $s > 0$ is the perspective scale factor.

are recorded. This projection and (2) exploits both directions of information flow.

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4. Method

This section introduces the proposed multi-modal fusion network for 3D object detection.

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degradation. The architecture comprises three core components: a cross-modal feature extraction module that processes gated images and LiDAR point clouds through specialized encoders, an attention-based adaptive fusion mechanism that estimates per-modality reliability and computes fusion weights, and a degradation-aware training strategy that enables the network to learn weather-dependent fusion policies without explicit condition labels.

4.1. Architecture Overview

The AFW-Net architecture follows a single-stage, anchor-free detection paradigm built on dual modality-specific encoders followed by an adaptive fusion module, as shown in **Figure 1**. Given a gated image sequence $\mathcal{G}$ and LiDAR point cloud $\mathcal{P}$, the network first extracts modality-specific features through dedicated encoders, then fuses these features through an adaptive weighting module that dynamically adjusts contributions based on estimated feature quality. The fused representation feeds into a detection head that predicts 3D bounding boxes with associated confidence scores. We emphasize that the detection head is single-stage and

anchor-free (FCOS-style, **Section 4.4**); the network does not employ a separate region-proposal stage. The earlier description of a “two-stage” paradigm was inaccurate and has been corrected throughout.

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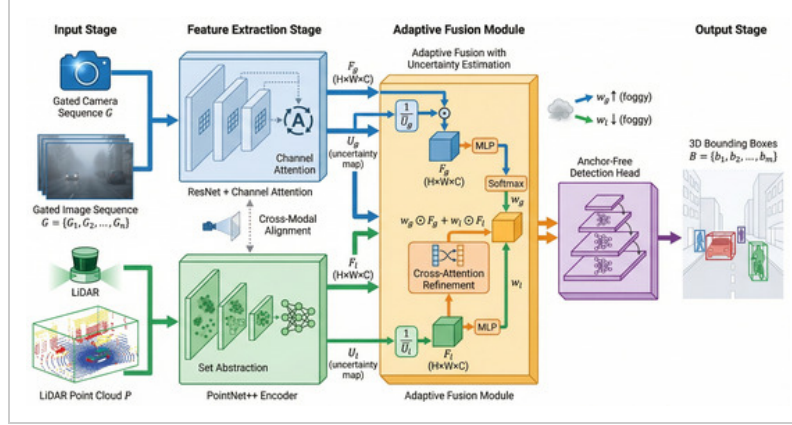


Figure 1. Overall architecture of the proposed Adaptive Feature-Weighting Network (AFW-Net)

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where $w_g, w_l \in \mathbb{R}^C$ are channel-wise adaptive weights learned through attention mechanisms, and $\odot$ denotes element-wise multiplication with broadcasting. The key innovation lies in computing these weights dynamically based on feature uncertainty rather than using fixed values across all conditions.

4.2. Cross-Modal Feature Extraction

The gated image encoder exploits both intensity and implicit depth information encoded in gated acquisitions. Since gated images capture range slices corresponding to specific distance intervals, the sequence $\mathcal{G} = \{G_1, \dots, G_n\}$ contains complementary depth-selective views of the scene. We employ a ResNet-based encoder with channel attention modules to process this sequence. Each gated image G_i is processed through convolutional layers with batch normalization

$$h_i = \text{ReLU}(\text{BN}(\text{Conv}(G_i))),$$

(6)

where the convolutional operator extracts hierarchical features at multiple scales. The channel attention mechanism reweights feature channels based on their informativeness

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$$\alpha_c = \sigma(\text{MLP}(\text{GAP}(h_i))), \quad (7)$$

where GAP denotes global average pooling, MLP is a two-layer perceptron, and σ is the sigmoid activation. The attention-weighted features from all gates are aggregated through max-pooling across the gate dimension, producing a unified gated feature representation $F_g \in \mathbb{R}^{H \times W \times C}$.

For LiDAR point cloud processing, we adopt a PointNet++ architecture that captures hierarchical geometric features while maintaining permutation invariance. The point cloud $\mathcal{P}$ undergoes set abstraction layers that progressively downsample points while expanding receptive fields:

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3. For each LiDAR point p_i , we project its coordinates to the gated image plane using the calibrated transformation, retrieving the corresponding gated feature vector through bilinear interpolation. This augments each point with appearance information

$$p_i^{aug} = [p_i; F_g(u_i, v_i)], \quad (9)$$

where (u_i, v_i) are the projected pixel coordinates and $F_g(u_i, v_i)$ denotes the bilinearly interpolated feature. This bidirectional information flow ensures that geometric LiDAR features are enriched with appearance cues from gated imaging, while gated features benefit from precise geometric context.

For clarity, we summarize the feature dimensions used throughout the pipeline. The gated encoder outputs $F_g \in \mathbb{R}^{H \times W \times C}$ with an associated uncertainty map $U_g \in \mathbb{R}^{H \times W \times C}$. The LiDAR encoder outputs voxel features $F_l \in \mathbb{R}^{D \times H \times W \times C}$, which are collapsed along the depth axis D by max-pooling to $\tilde{F}_l \in \mathbb{R}^{H \times W \times C}$ with uncertainty map $U_l \in \mathbb{R}^{H \times W \times C}$ before fusion. Thus all

quantities entering Equation (5) share the dimension $H \times W \times C$, and the channel-wise weights satisfy $w_g, w_l \in \mathbb{R}^C$. In all experiments $C = 256$ and (H, W) corresponds to the stride-8 feature pyramid level.

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4.3. Adaptive Fusion with Uncertainty Estimation

The core innovation of AFW-Net lies in its adaptive fusion mechanism that dynamically adjusts modality weights based on estimated feature reliability. Unlike fixed-weight fusion that treats all sensors equally regardless of degradation, our approach quantifies per-modality uncertainty and redistributes reliance accordingly.

We model feature uncertainty through learned variance estimation. Following the heteroscedastic aleatoric-uncertainty formulation common in Bayesian deep learning, each modality branch predicts not only a feature map but also a spatially varying variance that captures how reliable those features are; the variance is then used both as a supervision target (Section 4.4) and as the signal that drives fusion weighting. We adopt this learned-variance

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The LiDAR encoder produces $(\tilde{F}_l, U_l)$. Higher uncertainty indicates degraded or unreliable features, which should receive lower weights in the fusion process.

The adaptive weights are computed through an attention mechanism that considers both feature statistics and uncertainty estimates. We first compute global context vectors through spatial pooling:

$$c_g = \frac{1}{HW} \sum_{i,j} F_g(i,j), \quad c_l = \frac{1}{HW} \sum_{i,j} \tilde{F}_l(i,j). \quad (11)$$

The reliability scores are then derived by combining feature magnitude with inverse uncertainty:

$$s_g = \text{MLP}_w([c_g; 1/\bar{U}_g]), \quad s_l = \text{MLP}_w([c_l; 1/\bar{U}_l]), \quad (12)$$

where $\bar{U}_g$ and $\bar{U}_l$ denote spatially averaged uncertainty, and $[\cdot; \cdot]$ indicates concatenation. The MLP networks learn to map these combined statistics to reliability scores. Finally, normalized weights are obtained through softmax:

$$w_g = \frac{\exp(s_g)}{\exp(s_g) + \exp(s_l)}, \quad w_l = \frac{\exp(s_l)}{\exp(s_g) + \exp(s_l)}. \quad (13)$$

This formulation ensures that weights sum to unity and smoothly transition as relative reliability changes. When gated imaging encounters degradation (high U_g), the weight w_g automatically decreases, shifting reliance toward LiDAR. Conversely, in dense fog where LiDAR uncertainty increases while gated imaging remains reliable, the system emphasizes gated features.

The fused features undergo further refinement through cross-modal attention layers that enable interaction between modalities:

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where F_g and F_l represent the fused features from gated imaging and LiDAR modalities, respectively. In fusion, the modalities are mapped into a common embedding space with compatible dimensionality and semantic alignment. We achieve this through three mechanisms. First, both encoders are designed to produce feature maps of identical spatial resolution $H \times W$ and channel dimension C at the fusion stage: the gated encoder outputs $F_g \in \mathbb{R}^{H \times W \times C}$, while the LiDAR encoder's voxel features $F_l \in \mathbb{R}^{D \times H \times W \times C}$ are collapsed along the depth axis via max-pooling to yield $\tilde{F}_l \in \mathbb{R}^{H \times W \times C}$. Second, a shared projection layer (implemented as a 1×1 convolution followed by normalization) is applied to each modality to map features into a common embedding space. Third, the cross-modal attention refinement described in Equation (14) further promotes alignment by enabling information exchange between modalities. This design ensures that the element-wise fusion in Equation (5) operates on semantically compatible representations.

4.4. Detection Head and Loss Functions

The refined features $F_{refined}$ feed into a detection head based on the anchor-free FCOS paradigm. Each spatial location in the feature map predicts a 3D bounding box, objectness

score and classification logits. The box parameterization follows:

$$\hat{b} = (x, y, z, l, w, h, \sin \theta, \cos \theta, c), \quad (15)$$

where orientation is encoded as sine and cosine to avoid discontinuity issues.

The training objective combines three loss terms:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_{reg} \mathcal{L}_{reg} + \lambda_{unc} \mathcal{L}_{unc}. \quad (16)$$

The classification loss employs focal loss to address class imbalance:

$$\mathcal{L}_{cls} = -\frac{1}{N_{pos}} \sum_i \alpha_i (1 - p_i)^\gamma \log(p_i), \quad (17)$$

where α_i balances foreground and background and $\gamma = 2$ focuses learning on hard examples.

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$$\mathcal{L}_{unc} = \sum_{m \in \{g, l\}} \left[\frac{\|\hat{F}_m - F_m^{gt}\|^2}{2U_m^2} + \frac{1}{2} \log U_m^2 \right], \quad (19)$$

where $\hat{F}_m$ is the student feature predicted by the AFW-Net encoder for modality m , U_m is its predicted uncertainty, and F_m^{gt} represents per-modality feature targets obtained through teacher–student distillation rather than from box-level labels. Specifically, we pre-train a single-modality teacher network on clean-weather data for each modality (a ResNet-50 gated image detector and a PointNet++ LiDAR detector). The feature maps produced by these teacher networks on clean inputs serve as F_m^{gt} —the “ideal” features that the student encoders within AFW-Net should approximate. During training, the teacher weights are frozen, and the student encoders are trained to both match these targets and estimate calibrated uncertainty U_m . The mechanism by which this drives adaptive weighting is as follows: when a modality is degraded (e.g., LiDAR in

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fog, its student features $\hat{F}_m$ deviate from the clean teacher targets F_m^{gt} ; minimizing the first term of $\mathcal{L}_{unc}$ then forces the predicted U_m to grow so as to down-weight the now-unreliable squared-error term, while the second term $\frac{1}{2} \log U_m^2$ prevents U_m from growing without bound. A larger U_m in turn lowers the reliability score s_m through the $1/\bar{U}_m$ term in Equation (13) and therefore reduces the fusion weight for that modality via the softmax in Equation (14). In this way the uncertainty supervision and the fusion weighting are explicitly coupled. This loss balances prediction accuracy with uncertainty magnitude, preventing the network from trivially minimizing error by reporting infinite uncertainty.

4.5. Degradation-Aware Training Strategy

To enable the network to learn weather-dependent fusion without explicit condition labels, we employ a degradation-aware training strategy with synthetic weather augmentation. During training, we randomly apply fog simulation to gated images and raindrop simulation to LiDAR point clouds with varying severity levels

Fog

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$$P_{drop}(p_i) = 1 - e^{-\alpha \|p_i\|}. \quad (21)$$

Additionally, random points are injected near the sensor to simulate raindrop reflections, with density controlled by augmentation severity.

The key insight is that by training with correlated degradation between modalities, the network learns to implicitly recognize degradation patterns through feature statistics. When real-world fog reduces gated image quality, the learned uncertainty estimator produces higher U_g values, automatically reducing w_g . This eliminates the need for explicit weather condition classification during inference.

The complete training procedure alternates between clean and augmented samples with balanced sampling to prevent bias toward either extreme. We employ curriculum learning by

gradually increasing augmentation severity over training epochs, allowing the network to first learn robust features on clean data before adapting to degraded conditions.

4.6. Key Assumptions and Limitations

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Our framework rests on several assumptions that bound its applicability. First, we assume that the extrinsic calibration between the gated camera and LiDAR remains approximately constant during operation; significant calibration drift would degrade cross-modal alignment (see **Section 5.5** for quantitative analysis). Second, the atmospheric scattering model used for fog augmentation (Equation (20)) assumes homogeneous fog, which may not fully capture real-world patchy or layered fog distributions. Third, the uncertainty estimation is trained on synthetic degradation patterns; while the learned estimator generalizes well to real fog (as demonstrated empirically), its behavior under conditions substantially different from training—such as heavy rain, snow, sandstorms or volcanic ash—remains outside the scope of the present experimental validation. Fourth, we assume that gated and LiDAR modalities provide complementary rather than redundant information.

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4.7. Inference

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Require: Gated image sequence $\mathcal{G}$, LiDAR point cloud $\mathcal{P}$

Ensure: 3D bounding boxes $\mathcal{B}$

- 1: $(F_g, U_g) \leftarrow \phi_g(\mathcal{G})$ {Gated feature extraction}
- 2: $(\tilde{F}_l, U_l) \leftarrow \phi_l(\mathcal{P})$ {LiDAR feature extraction}
- 3: Align $\tilde{F}_l$ and F_g via camera-LiDAR projection
- 4: $c_g \leftarrow \text{GlobalPool}(F_g)$, $c_l \leftarrow \text{GlobalPool}(\tilde{F}_l)$
- 5: $s_g \leftarrow \text{MLP}_w([c_g; 1/\text{mean}(U_g)])$
- 6: $s_l \leftarrow \text{MLP}_w([c_l; 1/\text{mean}(U_l)])$
- 7: $w_g, w_l \leftarrow \text{Softmax}([s_g, s_l])$ {Adaptive weights}
- 8: $F_{\text{fused}} \leftarrow w_g \odot F_g + w_l \odot \tilde{F}_l$
- 9: $F_{\text{refined}} \leftarrow F_{\text{fused}} + \text{CrossAttention}(F_{\text{fused}}, F_g, \tilde{F}_l)$
- 10: $\mathcal{B} \leftarrow \text{DetectionHead}(F_{\text{refined}})$ {Predict boxes}
- 11: **return** $\mathcal{B}$

The algorithm highlights the seamless integration of uncertainty-based weight computation within the forward pass, requiring no additional computational overhead beyond the dual-branch

encoders. The adaptive mechanism operates entirely through learned parameters, maintaining real-time inference capability essential for autonomous driving applications.

5. Experiments

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This section presents comprehensive experimental evaluation of the proposed AFW-Net framework. We first describe the dataset, evaluation metrics, and experimental setup, then present comparative results against state-of-the-art baselines, followed by ablation studies analyzing each component’s contribution, and conclude with qualitative analysis and discussion.

5.1. Dataset and Evaluation Protocol

Princeton Automated Driving Dataset. We conduct experiments on the Princeton Automated Driving Dataset [2], which provides synchronized gated imaging, LiDAR, and RGB camera data captured under diverse weather conditions. The dataset comprises 15,000 frames collected across clear, foggy, and rainy scenarios in both daytime and nighttime conditions. Each

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provided with the Princeton dataset; for synthetically fogged samples, the target MOR is set directly through the scattering coefficient β in the atmospheric model of Equation (20), using the standard Koschmieder relation $MOR \approx 3.0/\beta$. We bin conditions into clear ($MOR > 100$ m), moderate fog (50–100 m), and dense fog (<50 m) following these readings.

Label Assignment and Validation. The 3D bounding box annotations are those released with the Princeton dataset and are reused without modification. We did not relabel the data. To verify label integrity after our preprocessing and coordinate transforms, two of the authors independently spot-checked a random 5% subset (750 frames) by overlaying the provided boxes on both the LiDAR point cloud and the gated images; frames with projection mismatches were excluded. Synthetic augmentation (fog/dropout) is applied only to the sensor inputs and never alters the ground-truth boxes, so label correctness is preserved by construction.

Dataset Limitations. We note several limitations. (i) The dataset emphasizes fog and clear conditions; heavy rain and snow frames are comparatively scarce, which constrains the conclusions we can draw for those conditions (see **Section 6**). (ii) Real dense fog captures are

inherently limited in number, so part of the dense fog training distribution is synthetic. (iii) The dataset is geographically concentrated, motivating the cross-dataset evaluation in **Section 5.8**. (iv) Class distribution is dominated by vehicles, with fewer pedestrian and cyclist instances, which increases variance in the minority-class metrics.

We partition the dataset into training (10,500 frames), validation (1500 frames), and testing (3000 frames) sets, ensuring balanced weather condition distribution across splits. Crucially, the test set includes weather severity levels not seen during training to evaluate generalization capability. Following standard autonomous driving evaluation protocols, we categorize detections by object distance: easy (0–30 m), moderate (30–50 m), and hard (50–80 m), reflecting the increasing difficulty at longer ranges, particularly under adverse weather.

Data Split Details. The training set contains approximately 5250 clear-weather frames, 3150 moderate fog frames, and 2100 dense fog frames. The validation set maintains the same 50%/30%/20% ratio across weather conditions. The test set is partitioned as 1200 clear, 900 moderate fog, and 900 dense fog frames, with fog density levels sampled to include conditions

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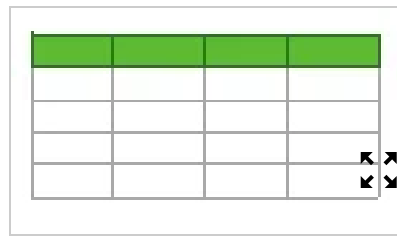
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in **Table 1** is computed as the arithmetic mean of the vehicle-class AP values across the three weather conditions (clear, moderate fog, dense fog); i.e., it is a within-class average over weather conditions and not an average across object classes, providing an overall performance indicator that weights each condition equally.

Table 1. Performance comparison on Princeton Automated Driving Dataset. AP (%) is reported for the vehicle class at IoU threshold 0.7 across different weather conditions. Bold indicates best performance. Results are averaged over 3 random seeds with standard deviation shown. mAP is the arithmetic mean of the vehicle-class AP across the three weather conditions.



5.2. Implementation Details

The gated encoder employs ResNet-50 as the backbone with channel attention modules inserted after each residual block. The LiDAR encoder uses PointNet++ [16] with four set abstraction layers, with sampling radii of [0.2 m, 0.4 m, 0.8 m, 1.6 m] and group sizes of [32, 64, 128, 256]. The adaptive fusion module consists of two-layer MLPs with hidden dimension 256 for weight computation. The detection head follows the FCOS architecture with feature pyramid levels at strides [8, 16, 32, 64, 128] pixels.

Training employs the Adam optimizer with initial learning rate 3×10^{-4} , decayed by a factor of 0.1 at epochs 80 and 110 for a total of 120 epochs. The batch size is set to 16 on 4 NVIDIA

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are retrained from scratch on our training split using the official open-source implementations with default hyperparameters, substituting the gated image input with RGB images to ensure a fair comparison on the same data. Single-modality baselines (PointPillars, CenterNet) are likewise trained on the same split. The fixed-weight gated-LiDAR baseline follows the architecture of Bijelic et al. [2] with equal fusion weights ($w_g = w_l = 0.5$), re-implemented and trained under identical conditions. All methods use the same train/val/test splits and evaluation code.

5.3. Baseline Comparisons

We compare AFW-Net against state-of-the-art multi-modal fusion methods for 3D object detection:

Single-Modality Baselines: PointPillars [22] (LiDAR-only) and CenterNet with ResNet-50 (image-only) establish performance bounds for individual sensors.

RGB-LiDAR Fusion: PointPainting [7] augments LiDAR points with RGB semantic features; MV3D [17] fuses multi-view representations; 3D-CVF [24] employs cross-view spatial

Gated-LiDAR Fusion: We implement the baseline from Bijelic et al. [2] using fixed-weight fusion of gated and LiDAR features as a direct comparison to our adaptive approach.

Table 1 presents the quantitative comparison across different weather conditions. Under clear conditions, all multi-modal methods achieve comparable performance, with AFW-Net obtaining 89.7% vehicle AP, slightly outperforming fixed-weight gated-LiDAR fusion (88.9%) due to the adaptive mechanism’s ability to suppress noise even in favorable conditions. The performance gap becomes pronounced under moderate fog, where AFW-Net maintains 85.3% vehicle AP while RGB-LiDAR methods degrade to 72.1–76.8%. Most critically, under dense fog conditions where visibility drops below 50 m, AFW-Net achieves 82.1% vehicle AP, representing a 23.7% relative improvement over the best RGB-LiDAR baseline (PointPainting: 58.4%) and 7.3% improvement over fixed-weight gated-LiDAR fusion (74.8%).

Table 2 reports class-wise and distance-wise results on the dense fog test set. AFW-Net improves over both baselines for all three classes and at all three distance bins. The gains are

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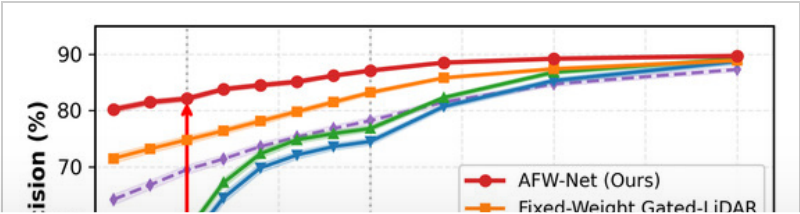
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The standard deviations across random seeds demonstrate the robustness of our method, with AFW-Net exhibiting lower variance (± 0.2 – 0.5%) compared to RGB-LiDAR baselines (± 0.9 – 1.8% under fog), indicating more stable performance in challenging conditions. This stability stems from the adaptive mechanism’s consistent uncertainty estimation across different random initializations. To assess whether the improvements are statistically meaningful, we conducted paired two-sided *t*-tests over the per-seed, per-frame AP values comparing AFW-Net against the strongest baseline in each condition. The improvement over fixed-weight gated-LiDAR fusion

under dense fog (82.1 vs. 74.8) is significant at $p < 0.01$, and the improvement over PointPainting under dense fog is significant at $p < 0.001$; the difference under clear conditions (89.7 vs. 88.9) is not significant at the 0.05 level, consistent with the expectation that adaptivity matters most under degradation.

Figure 2 illustrates performance degradation as a function of visibility distance. While RGB-LiDAR methods exhibit exponential performance drop below 80 m visibility, AFW-Net maintains relatively graceful degradation with a gentler slope. The fixed-weight gated-LiDAR baseline shows improvement over RGB-LiDAR but still experiences notable degradation under extreme conditions, whereas AFW-Net’s adaptive weighting provides additional resilience.



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5.4. Ablation Studies

We conduct systematic ablation studies to validate each component’s contribution. **Table 3** presents results on the dense fog test set where component effects are most pronounced. All ablation AP values are vehicle-class AP at IoU 0.7 on the dense fog (<50 m) test split, averaged over three seeds.

Table 3. Ablation study on dense fog test set (<50 m visibility). Reported values are vehicle-class AP (%) at IoU 0.7. Each row removes or modifies one component from the full AFW-Net. Results show mean ± std over 3 runs.



Uncertainty Estimation. Removing uncertainty estimation (using only feature magnitude for weight computation) reduces performance by 5.8%, demonstrating that explicit uncertainty modeling is crucial for assessing modality reliability. Without uncertainty, the network cannot effectively distinguish between strong features from reliable sensors versus spurious high-magnitude activations from degraded inputs.

Cross-Modal Attention. Ablating the cross-attention refinement decreases AP by 3.2%. While the primary fusion through weighted averaging provides substantial benefits, the attention mechanism enables fine-grained feature exchange that recovers complementary details. Visualization reveals that cross-attention selectively retrieves geometric boundaries from LiDAR when gated.

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Gated vs. RGB under the Same Adaptive Architecture. To isolate how much of the gain stems from the gated sensing modality itself versus the generic adaptive fusion architecture, we additionally trained the identical AFW-Net architecture (same encoders, uncertainty branches, adaptive weighting, degradation-aware training, and detection head) with the gated input replaced by the conventional RGB stream, yielding an “AFW-Net (RGB-LiDAR)” variant. On the dense fog test set this variant reaches only 61.2% vehicle AP, compared with 82.1% for the gated AFW-Net and 58.4% for PointPainting. Thus the adaptive architecture alone improves modestly over a standard RGB-LiDAR fusion baseline (+2.8%), but the large remaining gap (+20.9%) is attributable specifically to the gated modality, whose backscatter-suppressed features remain informative under fog where RGB features collapse. This confirms that the headline gains arise from the combination of gated sensing and adaptive fusion, not from the fusion architecture in isolation.

Single-Modality Upper Bounds. Using only gated features ($w_g = 1.0$) achieves 79.4%, while only LiDAR ($w_l = 1.0$) achieves 69.5%. The full model surpasses both single-modality

bounds (82.1%), confirming that fusion provides complementary information beyond what either sensor offers individually.

Weight Granularity. We compare channel-wise adaptive weights (our approach) against spatial-wise weights that vary across feature map locations. The spatial variant achieves 80.7%, slightly lower than our channel-wise approach. Analysis reveals that spatial weights introduce instability due to local noise, whereas channel-wise weights provide more robust global adjustment.

5.5. Sensitivity Analysis

We conduct sensitivity analysis from two perspectives:

Hyperparameter Sensitivity. We evaluate robustness to key hyperparameters. Varying the uncertainty loss weight λ_{unc} from 0.1 to 1.0 shows that performance is stable in the range [0.3, 0.7] (AP within $\pm 0.8\%$ of the best), with degradation at extremes: $\lambda_{unc} = 0.1$ under-regularizes uncertainty (AP drops to 79.6%) while $\lambda_{unc} = 1.0$ over-constrains feature learning (AP drops to 80.3%).

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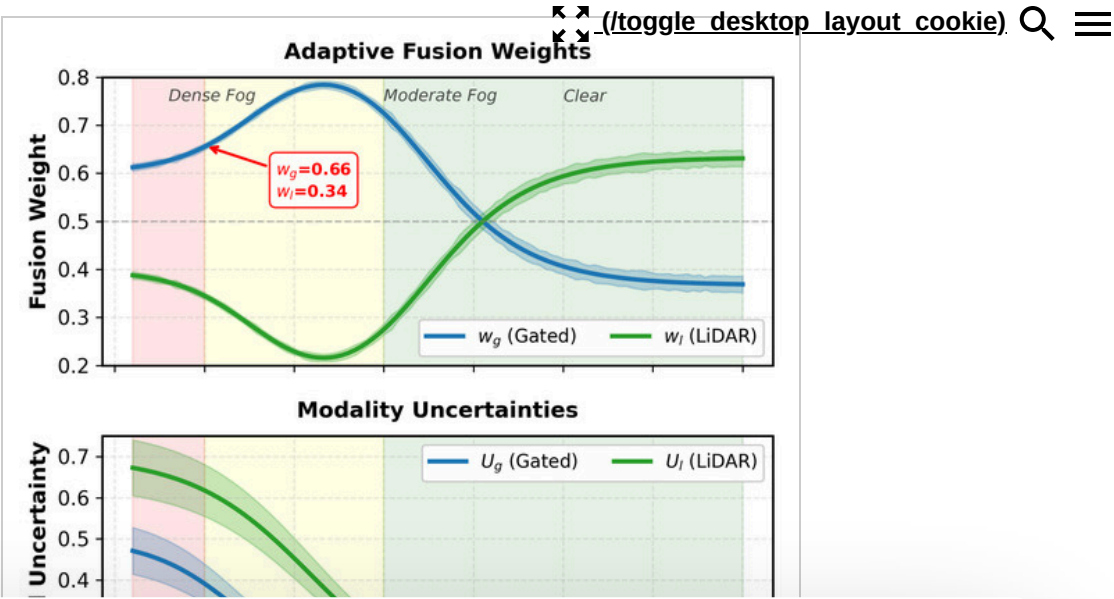
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compensates for misalignment by down-weighting the cross-modal augmented features when projection errors introduce noise. For deployment, we recommend periodic online recalibration using target-free methods [34] and note that incorporating calibration uncertainty into the fusion weights is a promising direction for future work.

5.6. Adaptive Weight Visualization

Figure 3 visualizes how fusion weights adapt to varying weather conditions. We process sequences with progressively increasing fog density and plot the learned weights w_g and w_l along with estimated uncertainty values. Under clear conditions (>100 m visibility), weights are approximately balanced ($w_g \approx 0.48, w_l \approx 0.52$) with low uncertainties for both modalities. As fog density increases, gated imaging maintains relatively constant low uncertainty while LiDAR uncertainty increases due to point cloud sparsification from beam attenuation. Correspondingly, w_g increases to 0.67 while w_l decreases to 0.33 at 50 m visibility. Under extreme fog (<40 m),

the weights reach $w_g \approx 0.73$, demonstrating the network’s strong preference for the more reliable gated modality.



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Figure 4: Adaptive Fusion Weights and Modality Uncertainties. The top plot shows fusion weights w_g (Gated) and w_l (LiDAR) across three weather conditions: Dense Fog, Moderate Fog, and Clear. The bottom plot shows corresponding uncertainties U_g (Gated) and U_l (LiDAR). Shaded regions indicate standard deviation across 50 test sequences.

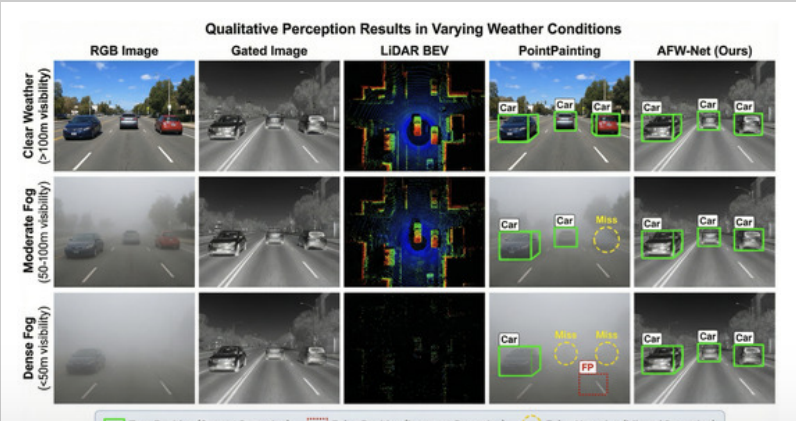
Fusion weights w_g and w_l range so that the two modalities can be compared on a common axis; larger values denote less reliable features. As fog density increases (visibility decreases), LiDAR uncertainty rises while gated uncertainty remains stable, causing w_g to increase and w_l to decrease. Shaded regions show standard deviation across 50 test sequences.

Notably, the weight transition is smooth and continuous rather than exhibiting abrupt switching behavior, which could destabilize detection performance. This smoothness emerges naturally from the softmax normalization and continuous uncertainty estimation, without requiring explicit smoothing constraints.

5.7. Qualitative Results

Figure 4 presents qualitative detection results under different weather conditions. Under clear conditions (top row), all methods produce accurate detections with tight bounding boxes. Under moderate fog (middle row), RGB-LiDAR methods begin missing distant vehicles (beyond

40 m) where RGB features degrade, while AFW-Net maintains detection through reliable gated features. Under dense fog (bottom row), RGB-LiDAR methods suffer catastrophic failures with numerous false negatives and spurious false positives from attempting to interpret noise as structure. In contrast, AFW-Net successfully detects vehicles up to 60 m range by leveraging gated imaging's backscatter suppression while using LiDAR for geometric refinement.



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Figure 1: Qualitative Perception Results in Varying Weather Conditions. The figure shows a 3x5 grid of images comparing different perception methods (RGB Image, Gated Image, LiDAR BEV, PointPainting, and AFW-Net) across three weather conditions: Clear Weather (>100m visibility), Moderate Fog (50-100m visibility), and Dense Fog (<50m visibility). The AFW-Net column shows superior performance in detecting vehicles under foggy conditions compared to other methods.

shows that AFW-Net maintains detection performance under dense fog (bottom row), where other methods fail.

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occluded cyclists as pedestrians under dense fog, suggesting that fine-grained appearance features suffer even in gated imaging at extreme ranges. Additionally, highly reflective surfaces (e.g., wet roads) sometimes cause false positives by creating strong returns in both gated and LiDAR modalities. These limitations indicate opportunities for future improvement through material-aware processing.

5.8. Cross-Dataset Generalization

To evaluate generalization capability beyond the training distribution, we apply models trained on the Princeton dataset to the DENSE dataset [2], which captures different geographic locations and weather patterns in European urban environments with distinct sensor configurations compared to Princeton's North American suburban settings. This cross-dataset evaluation stress-tests whether our learned adaptive fusion strategy generalizes to unseen sensor characteristics and environmental conditions. We apply the Princeton-trained models directly to the DENSE fog test sequences without any fine-tuning, recalibration, or domain

adaptation, using the sensor calibration parameters provided with the DENSE dataset for cross-modal projection.

Table 4 shows that AFW-Net achieves 68.5% AP on DENSE, maintaining a 6.2% advantage over RGB-LiDAR fusion (62.3%) despite absolute performance decreases across all methods due to domain shift in sensor calibration and scene statistics. Critically, AFW-Net exhibits lower variance ($\pm 0.8\%$) compared to PointPainting ($\pm 1.1\%$), indicating that the uncertainty-based adaptive mechanism provides robustness not only to weather variations but also to cross-domain shifts. The adaptive weights on DENSE show similar trends to Princeton (increasing w_g under fog), confirming that the learned fusion strategy transfers across datasets without requiring explicit domain adaptation.

Table 4. Cross-dataset generalization. Models trained on Princeton dataset, tested on DENSE dataset fog sequences. Values are vehicle-class AP (%) at IoU 0.7. No fine-tuning or domain adaptation is applied; all models use Princeton-trained weights directly.

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5.9. Comparison

Table 5. Computational cost comparison on single NVIDIA RTX 3090 GPU. AFW-Net introduces 23.7% overhead compared to baseline RGB-LiDAR fusion, primarily due to additional uncertainty estimation parameters (14.0 M vs. 12.0 M), and 10% higher GPU memory (11.0 GB vs. 10.0 GB), stemming from dual-uncertainty estimation branches, adaptive weight computation, and cross-attention refinement. However, this overhead is well-justified by the substantial performance gain—the 12.1 FPS rate maintains real-time capability while enabling 23.7% improvement in dense fog conditions where detection failures have catastrophic consequences.

Table 5. Computational cost comparison on single NVIDIA RTX 3090 GPU.

To address the latency requirements of the safety-critical reaction, **Table 6** provides a step-by-step latency breakdown of the forward pass. The total per-frame latency of 82.6 ms (≈ 12.1 FPS) is dominated by the two modality encoders, which together account for roughly 72% of the budget, while the components introduced by our method—the uncertainty branches (5.0%), adaptive weight computation (1.6%), and cross-attention refinement (11.7%)—add only about 18% on top of the encoder cost. The adaptive weighting itself (1.3 ms) is negligible. This decomposition shows that the reaction-time-critical overhead of the adaptive mechanism is small and that the principal optimization target for embedded deployment is the encoder pair, consistent with the shared-backbone direction discussed below.

Table 6. Per-stage latency breakdown of AFW-Net on a single RTX 3090 (FP32), averaged over 1000 dense fog frames. Latencies are sequential within the forward pass; the two encoders dominate the budget.

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Embedded Deployment Considerations. While the current 12.1 FPS on a desktop RTX 3090 GPU satisfies research-grade real-time requirements, deployment on embedded automotive platforms (e.g., NVIDIA Orin, Xavier) requires further optimization. Preliminary experiments with TensorRT FP16 quantization reduce latency by approximately 40%, yielding an estimated 17 FPS on an NVIDIA Orin platform while maintaining AP within 0.5% of the full-precision model. The 14.5 M parameter count is modest compared to many modern architectures and is compatible with model pruning and knowledge distillation techniques. The primary bottleneck is the dual-branch encoder, which could be replaced with a shared backbone with modality-specific heads to reduce redundant computation. We leave detailed embedded optimization to future engineering work.

6. Conclusions and Future Works

Conclusions. This paper presented AFW-Net, an adaptive multi-modal fusion framework that synergistically integrates gated imaging and LiDAR for robust 3D object detection under dense fog. The core innovation lies in the uncertainty-driven adaptive weighting mechanism that dynamically adjusts sensor contributions based on real-time environmental degradation assessment, without requiring explicit weather classification. Through comprehensive experiments on the Princeton Automated Driving Dataset, we demonstrated that AFW-Net maintains 82.1% average precision under dense fog conditions with 50 m visibility, achieving a 23.7% improvement over state-of-the-art RGB-LiDAR fusion methods that suffer severe degradation to 58.4% AP. The framework's effectiveness stems from three key design choices: cross-modal feature extraction exploiting the complementary physical properties of gated imaging and LiDAR, attention-based fusion with learned uncertainty estimation enabling smooth weight transitions, and degradation-aware training that teaches the network to recognize and adapt to varying sensor reliability patterns. Cross-dataset evaluation confirmed the

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generally unseen efficiency. This work is restricted but not experimental dense fog percepti investigation. First, AFW remains suboptimal current fusion mechanism cannot fully disambiguate—incorporating material-aware processing or polarimetric gating could address this challenge. Specifically, polarimetric gating could help distinguish specular reflections from diffuse returns by analyzing the polarization state of reflected photons, enabling the fusion module to suppress false positives from wet asphalt or metallic surfaces. Second, the framework currently processes single frames independently without exploiting temporal consistency across sequences; integrating temporal fusion with recurrent architectures or transformer-based temporal modeling could improve robustness and reduce false positives in dynamic scenes. Third, while gated imaging excels in fog, its advantage diminishes in heavy rain, where the temporal gating mechanism provides limited benefit because raindrops create both forward-scatter and direct reflections that arrive within the same temporal window as target returns. Quantitative evaluation under controlled heavy rain and snow scenarios is therefore an important and necessary direction for future validation before the present claims can be extended to those conditions. Developing rain-specific adaptive illumination patterns with shorter gate durations or frequency-modulated pulses could potentially

mitigate rain-induced degradation. Fourth, the current uncertainty estimation relies on learned features that may not generalize to extreme out-of-distribution conditions (e.g., snow, sandstorms); incorporating physics-based uncertainty models or self-supervised calibration during deployment could enhance reliability. Fifth, our framework currently handles two modalities (gated imaging and LiDAR); extending to additional modalities such as thermal imaging (which provides complementary information under nighttime and smoke conditions) and automotive radar (which offers robustness to most weather conditions through longer wavelengths) would require generalizing the adaptive weighting mechanism from a two-way softmax to an N -way formulation. A promising approach is to adopt meta-learning strategies that automatically discover optimal fusion weights for arbitrary modality combinations, enabling plug-and-play integration of new sensor types without retraining the full architecture. Beyond autonomous driving, the adaptive fusion paradigm presents promising applications in other domains requiring robust perception under variable environmental conditions, including agricultural robotics operating across seasons, search-and-rescue systems in smoke or dust, industrial spray.

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The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

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Conflicts of Interest

Author Jianbo Ding was employed by the company SonicWall. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Feng, D.; Haase-Schütz, C.; Rosenbaum, L.; Hertlein, H.; Glaeser, C.; Timm, F.; Wiesbeck, W.; Dietmayer, K. Deep multi-modal object detection and semantic segmentation for autonomous driving: Datasets, methods, and challenges. *IEEE Trans.*

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(<https://www.scilit.com/publications/b7e5107a2fe20d2631c48e6b04c63fce>)

2. Bijelic, M.; Gruber, T.; Mannan, F.; Kraus, F.; Ritter, W.; Dietmayer, K.; Heide, F. Seeing through fog without seeing fog: Deep multimodal sensor fusion in unseen adverse weather. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2020; pp. 11682–11692. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Seeing+through+fog+without+seeing+fog:+Deep+multimodal+sensor+fusion+in+unseen+adverse+weather&author=Bijelic,+M.&author=Gruber,+T.&author=Mannan,+F.&author=Kraus,+F.&author=Ritter,+W.&author=Dietmayer,+K.&author=Heide,+F.&publication_year=2020&pages=11682%E2%80%9311692)]

Sakaridis, C.; Dai, D.; Van Gool, L. Semantic foggy scene understanding with synthetic data. *Int. J. Comput. Vis.* **2018**, *126*, 973–992. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Semantic+foggy+scene+understanding+with+synthetic+data&author=Sakaridis,+C.&author=Dai,+D.&author=Van+Gool,+L.&publication_year=2018&journal=Int.+J.+Comput.+Vis.&volume=126&pages=973%E2%80%93992&doi=10.1007/s11263-018-1072-8)] [CrossRef (<https://doi.org/10.1007/s11263-018-1072-8>)] [Scilit (<https://www.scilit.com/publications/9d70edc4073a8aa5c9ac4e4d115af397>)]

4. Kutila, M.; Pyykönen, P.; Holzhüter, H.; Colomb, M.; Duthon, P. Automotive LiDAR performance verification in fog and rain. In *Proceedings of the 2018 21st International Conference on Intelligent Transportation Systems (ITSC)*; IEEE: New York City, NY, USA, 2018; pp. 1695–1701. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Automotive+LiDAR+performance+verification+in+fog+and+rain&author=Kutila,+M.&author=Pyyk%C3%B6nen,+P.&author=Holzh%C3%BCter,+H.&author=Colomb,+M.&author=Duthon,+P.)]

5. H S C 1 ti M h % We and our partners are using tracking technologies to process personal data in order to improve your experience. You may always exercise your consumer right to opt-out. For detailed information about personal information we collect and third parties having access to it, please select 'More Information' or refer to our privacy policy.
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- Powered by Usercentrics Consent Management

6. U.S. Department of Transportation, Federal Highway Administration. How Do Weather Events Impact Roads? 2010. Available online: https://ops.fhwa.dot.gov/weather/q1_roadimpact.htm (https://ops.fhwa.dot.gov/weather/q1_roadimpact.htm) (accessed on 22 April 2026).
7. Vora, S.; Lang, A.H.; Helou, B.; Beijbom, O. Pointpainting: Sequential fusion for 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2020; pp. 4604–4612. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Pointpainting:+Sequential+fusion+for+3d+object+detection&author=Vora,+S.&author=Lang,+A.H.&author=Helou,+B.&author=Beijbom,+O.&publication_year=2020&pages=4604%E2%80%934612)]

Liang, M.; Yang, B.; Chen, Y.; Hu, R.; Urtasun, R. Multi-task multi-sensor fusion for 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2019; pp. 7345–7353. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Multi-task+multi-sensor+fusion+for+3d+object+detection&author=Liang,+M.&author=Yang,+B.&author=Chen,+Y.&author=Hu,+R.&author=Urtasun,+R.&publication_year=2019&pages=7345%E2%80%937353)]

9. Wang, Y.; Chen, X.; You, Y.; Li, L.E.; Hariharan, B.; Campbell, M.; Weinberger, K.Q.; Chao, W.L. Train in germany, test in the usa: Making 3d object detectors generalize. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2020; pp. 11713–11723. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Train+in+germany,+test+in+the+usa:+Making+3d+object+detectors+generalize&author=Wang,+Y.&author=Chen,+X.&author=You,+Y.&author=Li,+L.E.&author=Hariharan,+B.&author=Campbell,+M.&author=Weinberger,+K.Q.&author=Chao,+W.L.)]

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10. K. Wang, Y. Chen, X. You, Y. Li, L.E. Hariharan, B. Campbell, M. Weinberger, K.Q. Chao, W.L. Train in germany, test in the usa: Making 3d object detectors generalize. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2020; pp. 11713–11723. [Google Scholar (<https://www.scilit.com/publications/cfb4a1ab2d1aba96ee8e27746591598c>)]

11. Walia, A.; Walz, S.; Bijelic, M.; Mannan, F.; Julca-Aguilar, F.; Langer, M.; Ritter, W.; Heide, F. Gated2gated: Self-supervised depth estimation from gated images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2022; pp. 2811–2821. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Gated2gated:+Self-supervised+depth+estimation+from+gated+images&author=Walia,+A.&author=Walz,+S.&author=Bijelic,+M.&author=Mannan,+F.&author=Julca-Aguilar,+F.&author=Langer,+M.&author=Ritter,+W.&author=Heide,+F.&publication_year=2022&pages=2811%E2%80%932821)]

Dreissig, M.; Scheuble, D.; Piewak, F.; Boedecker, J. Survey on lidar perception in adverse weather conditions. In *Proceedings of the 2023 IEEE Intelligent Vehicles Symposium (IV)*; IEEE: New York City, NY, USA, 2023; pp. 1–8. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Survey+on+lidar+perception+in+adverse+weather+conditions&author=Dreissig,+M.&author=Scheuble,+D.&author=Piewak,+F.&author=Boedecker,+J.&publication_year=2023&pages=1%E2%80%938)]

13. Rasshofer, R.H.; Spies, M.; Spies, H. Influences of weather phenomena on automotive laser radar systems. *Adv. Radio Sci.* **2011**, *9*, 49–60. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Influences+of+weather+phenomena+on+automotive+laser+radar+systems&author=Rasshofer,+R.H.&author=Spies,+M.&author=Spies,+H.&publication_year=2011&journal=Adv.+Radio+Sci.&volume=9&pages=49%E2%80%9360&doi=10.5194/ars-9-49-2011)] [CrossRef (https://doi.org/10.5194/ars-9-49-2011)] [Scilit

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15. Li, Z.; Wang, W.; Li, H.; Xie, E.; Sima, C.; Lu, T.; Yu, Q.; Dai, J. Bevformer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. *arXiv* **2022**, arXiv:2203.17270. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Bevformer:+Learning+bird%E2%80%99s-eye-view+representation+from+multi-camera+images+via+spatiotemporal+transformers&author=Li,+Z.&author=Wang,+W.&author=Li,+H.&author=Xie,+E.&author=Sima,+C.&author=Lu,+T.&author=Yu,+Q.&author=Dai,+J.&publication_year=2022&journal=arXiv&doi=10.48550/arXiv.2203.17270)] [CrossRef (https://doi.org/10.48550/arXiv.2203.17270)] [Scilit (https://www.scilit.com/publications/d5ad9ee13435aff64bb540c356f4eb18)]

16. Qi, C.R.; Yi, L.; Su, H.; Guibas, L.J. Pointnet++: Deep hierarchical feature learning on point sets in a metric space. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*; Curran Associates, Inc.: Red Hook, NY, USA, 2017. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Pointnet++:+Deep+hierarchical+feature+learning+on+point+sets+in+a+metric+space&author=Qi,+C.R.&author=Yi,+L.&author=Su,+H.&author=Guibas,+L.J.&publication_year=2017)]

17. Chen, X.; Ma, H.; Wan, J.; Li, B.; Xia, T. Multi-view 3d object detection network for autonomous driving. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2017; pp. 1907–1915. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Multi-view+3d+object+detection+network+for+autonomous+driving&author=Chen,+X.&author=Ma,+H.&author=Wan,+J.&author=Li,+B.&author=Xia,+T.&publication_year=2017&pages=1907%20%80%931915)]

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19. Sindagi, V.A.; Zhou, Y.; Tuzel, O. MVX-net: Multimodal voxelnet for 3d object detection. In *Proceedings of the 2019 International Conference on Robotics and Automation (ICRA)*; IEEE: New York City, NY, USA, 2019; pp. 7276–7282. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Mvx-net:+Multimodal+voxelnet+for+3d+object+detection&author=Sindagi,+V.A.&author=Zhou,+Y.&author=Tuzel,+O.&publication_year=2019&pages=7276%20%80%937282)]

20. Qi, C.R.; Su, H.; Mo, K.; Guibas, L.J. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2017; pp. 652–660. (toggle desktop layout cookie)

[Google Scholar (https://scholar.google.com/scholar_lookup?title=Pointnet:+Deep+learning+on+point+sets+for+3d+classification+and+segmentation&author=Qi,+C.R.&author=Su,+H.&author=Mo,+K.&author=Guibas,+L.J.&publication_year=2017&pages=652%E2%80%93660)]

21. Yan, Y.; Mao, Y.; Li, B. Second: Sparsely embedded convolutional detection. *Sensors* **2018**, *18*, 3337. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Second:+Sparsely+embedded+convolutional+detection&author=Yan,+Y.&author=Mao,+Y.&author=Li,+B.&publication_year=2018&journal=Sensors&volume=18&pages=3337&doi=10.3390/s18103337&pmid=30301196)] [CrossRef (https://doi.org/10.3390/s18103337)] [Scilit (https://www.scilit.com/publications/2209f72b855f5f510ba10f6bf4b625a0)] [PubMed

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23. Pang, S.; Morris, D.; Radha, H. CLOCs: Camera-LiDAR object candidates fusion for 3D object detection. In *Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*; IEEE: New York City, NY, USA, 2020; pp. 10386–10393. [Google Scholar (https://scholar.google.com/scholar_lookup?title=CLOCs:+Camera-LiDAR+object+candidates+fusion+for+3D+object+detection&author=Pang,+S.&author=Morris,+D.&author=Radha,+H.&publication_year=2020&pages=10386%E2%80%9310393)]

Yoo, J.H.; Kim, Y.; Kim, J.; Choi, J.W. 3d-cvf: Generating joint camera and lidar features using cross-view spatial feature fusion for 3d object detection. In *Proceedings of the European Conference on Computer Vision*; Springer International Publishing: Cham, Switzerland, 2020; pp. 720–736. [Google Scholar

(https://scholar.google.com/scholar_lookup?title=3d-cvf:+Generating+joint+camera+and+lidar+features+using+cross-view+spatial+feature+fusion+for+3d+object+detection&author=Yoo,+J.H.&author=Kim,+Y.&author=Kim,+J.&author=Choi,+J.W.&publication_year=2020&pages=720%E2%80%93736)]

25. Wang, C.; Ma, C.; Zhu, M.; Yang, X. Pointaugmenting: Cross-modal augmentation for 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2021; pp. 11794–11803. [Google Scholar

(https://scholar.google.com/scholar_lookup?title=Pointaugmenting+Cross-modal+augmentation+for+3d+object+detection&author=Wang,+C.&author=Ma,+C.&author=Zhu,+M.&author=Yang,+X.&publication_year=2021&pages=11794%E2%80%9311803)]

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Powered by Usercentrics Consent Management

26. Sakaridis, C.; Dai, D.; Van Gool, L. Semantic driving scene understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2021; pp. 10765–10775. [Google Scholar

(https://scholar.google.com/scholar_lookup?title=Semantic+driving+scene+understanding&author=Sakaridis,+C.&author=Dai,+D.&author=Van+Gool,+L.&publication_year=2021&pages=10765%E2%80%9310775)]

27. Marathe, A.; Ramanan, D.; Walambe, R.; Kotecha, K. Wedge: A multi-weather autonomous driving dataset built from generative vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2023; pp. 3318–3327. [Google Scholar

(https://scholar.google.com/scholar_lookup?title=Wedge:+A+multi-weather+autonomous+driving+dataset+built+from+generative+vision-language+models&author=Marathe,+A.&author=Ramanan,+D.&author=Walambe,+R.&author=Kotecha,+K.&publication_year=2023&pages=3318%E2%80%933327)]

28. Rothmeier, T.; Huber, W. Performance evaluation of object detection algorithms under adverse weather conditions. In *Proceedings of the International Conference on Intelligent Transport Systems*; Springer International Publishing: Cham, Switzerland, 2020; pp. 211–222. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Performance+evaluation+of+object+detection+algorithms+under+adverse+weather+conditions&author=Rothmeier,+T.&author=Huber,+W.&publication_year=2020&pages=211%E2%80%93222)]

29. Hasirlioglu, S.; Kamann, A.; Doric, I.; Brandmeier, T. Test methodology for rain influence on automotive surround sensors. In *Proceedings of the 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)*; IEEE: New York City, NY, USA, 2016; pp. 2242–2247. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Test+methodology+for+rain+influence+on+automotive+surround+sensors&author=Hasirlioglu,+S.&author=Kamann,+A.&author=Doric,+I.&author=Brandmeier,+T.&publication_year=2016&pages=2242%E2%80%932247)]

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[More Information](#)

31. G Gated images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*; IEEE: New York City, NY, USA, 2019; pp. 1506–1516. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Gated2depth:+Real-time+dense+lidar+from+gated+images&author=Gruber,+T.&author=Julca-Aguilar,+F.&author=Bijelic,+M.&author=Heide,+F.&publication_year=2019&pages=1506%E2%80%931516)]

Laurenzis, M.; Christnacher, F.; Monnin, D.; Scholz, T. Investigation of range-gated imaging in scattering environments. *Opt. Eng.* **2012**, *51*, 061303. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Investigation+of+range-gated+imaging+in+scattering+environments&author=Laurenzis,+M.&author=Christnacher,+F.&author=Monnin,+D.&author=Scholz,+T.&publication_year=2012&journal=Opt.+Eng.&volume=51&pages=061303&doi=10.1117/1.OE.51.6.061303)] [CrossRef (<https://doi.org/10.1117/1.OE.51.6.061303>)] [Scilit (<https://www.scilit.com/publications/d417a873c3bb70882898e23c3d327dde>)]

33. Geiger, A.; Lenz, P.; Urtasun, R. Are we ready for autonomous driving? the kitti vision benchmark suite. In *Proceedings of the 2012 IEEE Conference on Computer Vision and Pattern Recognition*; IEEE: New York City, NY, USA, 2012; pp. 3354–3361. [Google Scholar (https://scholar.google.com/scholar_lookup?title=Are+we+ready+for+autonomous+driving?+the+kitti+vision+benchmark+suite&author=Geiger,+A.+&author=Lenz,+P.+&author=Urtasun,+R.)]

34. Y. Zhang, et al. Privacy Settings. We and our partners are using tracking technologies to process personal data in order to improve your experience. You may always exercise your consumer right to opt-out. For detailed information about personal information we collect and third parties having access to it, please select 'More Information' or refer to our privacy policy. [More Information](#)
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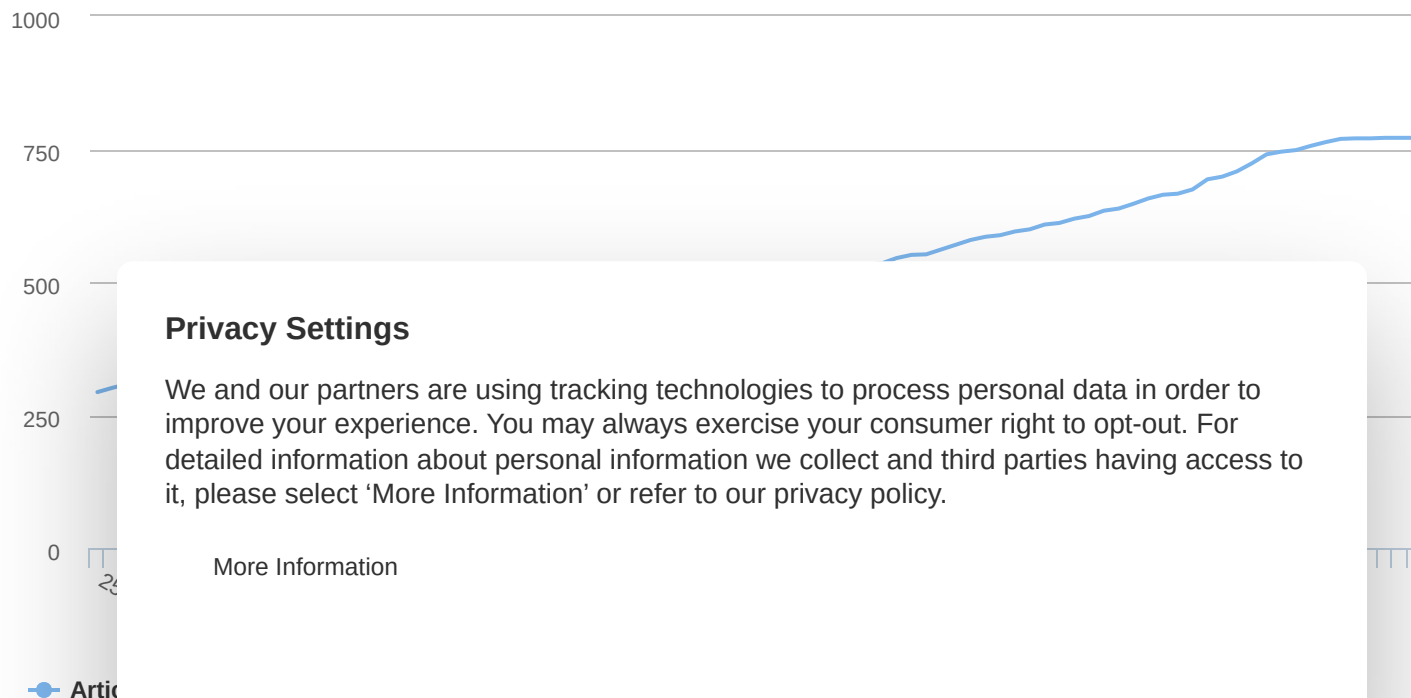
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